BioSpecInfo

Traceability Matrix — BioSpecInfo

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Traceability Matrix — BioSpecInfo

Field	Value
Software	BioSpecInfo
Author	Samuele Pio Provenzano
Version described	`bsi-v241`
Purpose	Connect every declared requirement to the implementation that realises it and to the bench that verifies it.

> **How to read this matrix.** Every row is a closed chain: a requirement, the point in the code that realises it, and the bench that verifies it works. A row without a bench is an **unverified** requirement and is declared as such, not hidden: the usefulness of a matrix lies precisely in making the gaps visible.
>
> The outcomes of the benches cited here are reported in full, with their complete output, in `../evidence/RAPPORTO-VERIFICA.md`, which is generated automatically.

1. Scientific requirements

ID	Requirement	Implementation	Verification bench
SCI-01	A drug structure must come from an external source and match that source, not merely weigh correctly	`FARM_DATA` in `index.html` — **263 entries**, 36 declaring a ChEMBL identifier and formula	`audit_farmaci` — the formula is **rebuilt by counting the graph's atoms** and compared with ChEMBL's: 36 of 36. Weight alone was not enough: three pre-existing entries (lenalidomide, palbociclib, aripiprazole) were **isomers** — same weight, different molecule — and were replaced. Two ratchets: verified provenance ≥ 36, readable structures ≥ 240
SCI-02	Functional groups must be recognised on the structure, not on the text of the SMILES	`bsi-spettri.js` §1, SMARTS patterns via RDKit	`test_spettri` — 8 reference molecules, additive recognition
SCI-03	A molecule with several functional groups must show them all	`bandeDaGruppi()`, additive construction	`test_spettri` §1 — acetylsalicylic acid: ester **and** acid

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SCI-04	Spectra must respect the conventions of their technique	`svgIR()`, `makeNMRsvg()`, `drawUVSpectrum()`, `drawMSSpectrum()`	`test_assi` (SVG), `test_assi_canvas` (canvas)
SCI-05	A computed spectrum must be reproducible	every `Math.random()` removed from the traces	`test_assi` §3, `test_spettri` §8 — byte-for-byte comparison
SCI-06	The height of ^{1}H peaks must represent the integration, or not be asserted at all	`makeNMRsvg()`; `_nmrSvg()` draws bands of uniform height	`test_assi` §4
SCI-07	Conjugation must act on the individual carbonyl	SMARTS `acidoConiug`, `estereConiug`, `chetoneConiug`, `aldeideConiug`	`test_spettri` §1 — aspirin 1760/1690; acetophenone 1690
SCI-08	An ambiguous physical constant must not be resolved arbitrarily	`costante_fisica` in `bsi-ai-hub.js` — collects every candidate	`test_costanti` — 45 checks, including the refusals for ambiguity
SCI-09	The limits of the predictors must be declared to the user	chart headers; `docs/06` §3.5	`verifica_guida` — consistency between promises and code
SCI-10	Molecular symmetry must determine the selection rules correctly	`ssimm` section in `index.html`	`test_simmetria` — 26 checks: normal modes, centre of inversion, mutual exclusion
SCI-12	A chromatogram must follow from the parameters, not be drawn by hand	`scroma` section: σ = t_R/√N, R_s from distance and widths	`audit_stabilita` (opening and drawing), inspection of the values against the definitions
SCI-13	A calibration line must declare the uncertainty of its coefficients and recognise the two ways it misleads	`staratura` section: ordinary least squares, F-test line⟷parabola, deleted residual	Verified on the three example sets included: linear (F = 0.3), non-linear (F = 186.6 > 7.71), with an outlier (1 detected)
SCI-14	A QSAR model must be evaluated on scaffolds never seen, not on analogues from the same set	`bsi-cheminfo.js` §9 — `divisionePerScaffold()` alongside `divisioneCasuale()`	`test_cheminfo` — the scaffold split leaves no scaffold shared between training and test

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SCI-15	A model score must be compared against chance, not presented on its own	`bsi-cheminfo.js` §9 — retraining on scrambled labels, eight repetitions	`test_cheminfo` — verified **in both directions**: on a learnable signal R^2 0.394 beats all eight twins (margin 0.287); on random labels R^2 −0.199 does **not** beat them
SCI-16	Two spellings of the same molecule must not be counted twice, nor land on opposite sides of the split	`bsi-cheminfo.js` §1 — deduplication on the canonical SMILES after the largest fragment	`test_cheminfo` — `OC(=O)C` and `CC(=O)O` collapse into one entry; salts lose the counterion
SCI-17	Similarity between molecules must be computed, not estimated	`bsi-cheminfo.js` §3 — Tanimoto and Dice over Morgan, RDKit, pattern and MACCS fingerprints	`test_cheminfo` — compared against hand-computable values, including the 0/0 = 1 convention
SCI-18	A validation on a single split, over a small set, is a noisy number: it must come with its dispersion	`bsi-cheminfo.js` §9-bis — `validazioneIncrociata()` over **scaffold-grouped** folds	`test_cheminfo` — the folds cover every molecule once, **no scaffold sits in two folds**, and the same seed gives the same folds
SCI-19	A result that changes on every run is verifiable by nobody	seeded mulberry32 generator for splits, scrambles and folds; the seed is written into the report	`test_cheminfo` — two calls with the same seed give identical folds; the method report declares the seed
SCI-20	Results must be able to leave the tool, and the method must leave with them	`bsi-cheminfo.js` §9-ter — `esportaCsv()` and `rapportoMetodo()`, both formed in memory in the browser	`test_cheminfo` — columns consistent with the header, a name containing a comma quoted, residual computed, empty cell for missing activity; the report carries seed, fingerprint, verdict, limits
SCI-21	A number presented as structural similarity must come from a fingerprint, and one implementation must compute it	`rdkit_lab.html` loads `bsi-cheminfo.js` and uses its `tanimoto()` and `fingerprint()`; the eight-bit similarity has been removed	`test_cheminfo` — page and engine give the same Tanimoto over all **28 pairs**, deviation **exactly 0**; the fake-similarity functions no longer exist; every reference structure is valid and carries the **monoisotopic mass** of the drug it names

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SCI-22	Conversions between molecular languages must be checked against an external source or against hand calculation, and coordinates written to a file must be real and declared	`bsi-molingue.js` — **seventeen outputs**, from SMILES to CML	`test_lingue` — 7 **literature InChI keys**, full SMILES → molfile → SMILES round-trip, 26 hand-written names, **7 required refusals**; mass composition against hand calculation (alanine C 40.44%, aspirin C 60.00%); XYZ and PDB with non-null coordinates **and** declared 2D
SCI-23	Angles, lengths and dihedrals are measured on the coordinates, and on flat coordinates they are not shown	dot product for angles, cross product for dihedrals	`test_mol3d` — tetrahedron **109.4712°**, water 104.47°, CO_2 180°, BF_3 120°, ammonia 106.13°; dihedrals 0°, 60°, 90°, 180°; ethanol C–C 1.509 Å; and the refusal tested in both directions
SCI-24	A spectrum read from a file must be decoded according to the format, and the peaks found must be the ones that are there — no more, no fewer	`bsi-spettrolettore.js` — JCAMP-DX with full ASDF (PMAI, DIF, DUP) and the integrity check the format prescribes; peaks by **prominence**, noise from MAD × 1.4826	`test_spettrolettore` — **61 checks**. On three Gaussians built at 1715, 2950, 3400 cm^{-1} under noise it finds three: 1716, 2952, 3402. **Both ways**: on pure noise it finds **zero**. ASDF verified symbol by symbol (`abcdefghi` = −1…−9; `n` = −5, not −4)
SCI-25	A band must be declared *compatible with*, never assigned to a single possibility	`BANDE_IR` — 24 ranges; `assegnaIR()` returns **all** compatible assignments	`test_spettrolettore` — at 1715 cm^{-1} both a ketone C=O and a carboxylic acid C=O appear; the section states that it does not deduce the structure
SCI-26	A published model must be checked against **its own** declared error, and a learned model against chance	`bsi-cheminfo.js` §10 — ESOL (Delaney 2004) with its uncertainty, four drug-likeness filters reported separately, random forest with out-of-bag error	`test_spettrolettore` §4-5 — ESOL against four experimental values, mean deviation **0.68**, inside the declared ~1 log unit; forest out-of-bag R^2 **0.90** on a non-linear relationship and **−0.16** on pure noise; deterministic for a given seed and different for another
SCI-27	A structure without a source must not get in, not even one that is known	collected from ChEMBL, `tools/dati/farmaci_v187.json` carries each entry's identifier	**Two refusals**: ivermectin (`structure_type NONE`, a mixture of homologues) and semaglutide (a protein, `SEQ`). And two entries kept under the **record's** name rather than the query's: «morphine» → apomorphine, «levothyroxine» → liothyronine

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SCI-28	A predicted spectrum must say WHICH atoms produce each signal, and the deviation must be measured on molecules that did not choose the parameters	`bsi-pretsch.js` — the estimation tables transcribed in full from the source (Pretsch 4th ed.: 91 ^{13}C and 66 ^{1}H rows for benzenes, 42 for ethylenes, 31 for alkanes, 24 for ^{13}C aliphatics, 4×4 steric corrections); `bsi-nmr.js` — the reasoning that applies them: ring position obtained by walking the cycle, additive scheme with a cyclic reference compound, chemical equivalence by shell environment code	`test_nmr` — **56 checks**. Two separate sets: tuning **0.71 ppm** over 9 molecules, **validation 0.92 ppm** over 22 never used for tuning, worst case **4.9 ppm**, ^{1}H **0.06 ppm** (0.03 on the aromatics alone). The bench requires validation to stay **worse** than tuning: were they equal, someone would have moved a molecule between the two sets
SCI-29	Chemically equivalent atoms must give ONE signal, not one per atom	environment code built as concentric shells by distance (the HOSE-code principle), with aromatic bonds written as such and not in Kekulé form	`test_nmr` — eight molecules whose signal count is known: benzene **1** (not six), toluene 5, p-xylene 3, naphthalene 3, biphenyl 4, aspirin 9. All correct
SCI-30	From an image one can recover the SHAPE of a spectrum, not its calibration	`bsi-digitalizza.js` — `bsi-spettrolettore.js` delegates `daImmagine()` to it, a single implementation. The axis frame is found first, the trace is extracted at **sub-pixel** resolution (darkness-weighted centroid), the run is chosen by continuity **and** by darkness, both axes are calibrated linearly or logarithmically, %T is converted to absorbance with $A = -\log_{10} T$	`test_digitalizza` — **45 checks** over seven constructed figures: mean deviation **2.2 cm⁻¹**, worst **7.3** (one column is worth ~4 cm⁻¹). **Both ways**: from a blank sheet and from the app icon no curve comes out, and the crude fallback on the same figure with frame and labels finds **one** peak instead of three. `test_spettrolettore` for the routing
SCI-31	A two-dimensional NMR map must say WHICH atoms produce each correlation, and must not connect what the topology does not connect	`bsi-nmr2d.js` — one-bond HSQC with inverted sign on CH_2, three-bond COSY between non-equivalent protons, two- and three-bond HMBC; exchangeables excluded from off-diagonal spots	`test_nmr2d` — **40 checks**, in both directions: hexafluorobenzene gives no HSQC spots, benzene **no** off-diagonal COSY spots (its six protons are equivalent), methane **no** HMBC spots despite having four protons
SCI-32	A program-built 3D geometry must carry THE SAME indices as the predictor, otherwise the peak↔atom link lights up the wrong atom	`bsi-geom3d.js` — distance geometry over the `bsi-nmr.js` graph, hydrogens appended at the end, generator seeded from the SMILES	`test_geom3d` — **29 checks**: every heavy atom has the same index and element across five molecules; bonds and angles within tolerance of literature values; **in both directions** benzene comes out planar and cyclohexane does not

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SCI-33	A document must be opened in full, and whatever could NOT be read must be declared — a reader that returns little without saying so makes the document look nearly empty	`bsi-documento.js` — PDF page by page with its text layer (PDF.js), images, text, Office archives opened with `DecompressionStream` and no libraries	`test_documento` — **47 checks**: PDF, Word and text genuinely opened and their content verified; **in both directions** an image declares there is no optical character recognition and produces no working, an unknown format is refused rather than guessed, a binary file named `.txt` is recognised as binary
SCI-34	Spectroscopic data recognised in a text must carry their PROVENANCE, and the working must declare what it is missing	`bsi-quesito.js` — recognition by section (IR, MS, ^{1}H, ^{13}C) with the boundaries between sections, then `bsi-elucida.js` for the step-by-step working	`test_documento` — formula, bands, masses with intensities, ^{1}H signals with integral and multiplicity, ^{13}C signals; **in both directions**: $CDCl_3$ does not become the compound, a band outside 400-4000 is not a band, without an MS label there are no masses, a new paragraph closes the section, a text with no spectra produces no data
SCI-35	A figure with a frame, labels and a grid must not be digitised as if it were only the curve: the axes and the numbers are dark pixels just like the trace	`bsi-digitalizza.js` — the frame is sought among the CONTINUOUS lines in the margins, and a border is told from a gridline by its **irregular spacing** relative to the family **or** by its thicker, darker stroke; the crop is moved past the line thickness; vertical gridline columns are marked as not measured and reconstructed from their neighbours	`test_digitalizza` §1-2 — on a figure with a frame **and** a grid (11 vertical families, 7 horizontal) it finds all four sides and the peaks stay within 2.2 cm^{-1}; **both ways**, on a grid-only figure it does **not** invent a frame and says so, and a **flat baseline** is not mistaken for an axis
SCI-36	A digitised curve must be able to leave the application, and the exported file must be valid	`bsi-digitalizza.js` — CSV output with the units in the header, and JCAMP-DX 4.24 in (XY..XY) form with FIRSTX/LASTX/NPOINTS and the provenance written inside the file	`test_digitalizza` §4 — the **full round trip**: the exported JCAMP is read back by `bsi-spettrolettore.js`'s `leggi()` and returns with the same number of points, the same end points and the same units
SCI-38	A quantum solver must reproduce the exact solutions that exist, not merely draw plausible curves	`bsi-fis-schrodinger.js` — finite differences with Sturm bisection and inverse iteration; Crank-Nicolson in time; Morse with literature parameters	`test_fisica` — harmonic oscillator, infinite well, numerical Morse = exact, norm conserved to 10^{-9}, exact transmission
SCI-39	An ab initio calculation must reproduce the published energies for the same basis and geometry	`bsi-fis-hartreefock.js` — RHF/STO-3G, McMurchie-Davidson integrals, Boys function, SCF with DIIS	`test_fisica` — Szabo-Ostlund H_2 (−1.116714 Eh) and HeH^+ (−2.860662), Crawford H_2O (−74.942079928) to 10^{-6}, nuclear repulsion, Mulliken charges summing to zero

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SCI-40	Computed physical-chemistry quantities (entropies, spectra, diffraction, decays) must match independent reference values	`bsi-fis-termostat.js`, `bsi-fis-rotovib.js`, `bsi-fis-cristalli.js`, `bsi-fis-nucleare.js`, `bsi-fis-dinamica.js`, `bsi-fis-matematica.js`	`test_fisica` — NIST-JANAF S° of eight gases, HCl r_e and B from combination differences, 2θ of the NaCl/Si/Cu/Fe reference patterns and absences, ^{99m}Tc peak, ^{14}C, ^{56}Fe B/A, energy conserved in molecular dynamics, Hückel, FFT, RK4, N_2O_5 E_a
SCI-41	A symbolic calculation must be exact or stay silent: a wrong antiderivative or derivative teaches the error with the face of a result	`bsi-mat-cas.js` — canonical form over exact fractions, derivatives with a log of the rules, antiderivatives returned **only** if their derivative matches the function at six points, Taylor series by truncated power-series arithmetic, limits by series and Richardson extrapolation	`test_matematica` — table derivatives and a complicated derivative against finite differences, antiderivatives by parts and by substitution, and **both ways**: e^(−x²) does **not** get an antiderivative; ∫e^(−x²) = √π, Taylor coefficients 1/120 and 17/315, standard limits,
SCI-42	Linear algebra must give exact results and the steps that produce them; balancing a reaction must come from conservation of atoms and charge	`bsi-mat-lineare.js` — Gauss-Jordan in BigInt fractions, Faddeev-LeVerrier characteristic polynomial, exact rational roots before Durand-Kerner; null space of the composition matrix	`test_matematica` — integer inverse of the Hilbert matrix and det 1/2160, a 4×4 det recomputed separately, eigenvalues of tridiag(−1, 2, −1), a defective matrix, permanganate, redox with charges, Wöhler; **both ways** $H_2 + O_2 \rightarrow H_2O + H_2O_2$ declared not unique
SCI-43	Laboratory statistics must reproduce the reference tables and examples, and uncertainty propagation must be checkable	`bsi-mat-statistica.js` — t and F from the incomplete beta function, Grubbs with the computed critical value, Rorabacher's Dixon, regression with s_y/x, LOD, LOQ and sample uncertainty; propagation with symbolic derivatives and Monte Carlo	`test_matematica` — t(0.975; 9), t(0.975; 4), F(0.975; 4, 4), Grubbs n = 5 and 10, the Miller & Miller calibration example (b, a, s_y/x, r, x_0 and $s(x_0)$), propagation matching a 20 000-draw Monte Carlo
SCI-44	An exam solver must produce the correct working, not just the result, and state when a step is numerical	`bsi-mat-risolutore.js` (domains, limits via Taylor, partial fractions, improper integrals, series with tests, two variables, extrema on compact sets, double integrals, differential forms, ODEs with Cauchy) and `bsi-fis-esame.js` (48 General Physics I and II models)	`test_esami` — results checked against SymPy, hand calculation or a different formula (range, Cramer, conservation laws, geostationary orbit); **every** solved-paper exercise produces a working; **both ways**: e^(−x²) has no antiderivative, $\int_1^{\infty} 1/x$ diverges, Σ 1/(n ln n) diverges

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SCI-45	Electrode kinetics, transition-state, photophysics, adsorption and polymer quantities must recover the parameters the data were generated from	`bsi-fis-elettrodi.js`, `bsi-fis-superfici.js`	`test_esami` — α and i_0 from Tafel, ΔH‡ and ΔS‡ from Eyring, n_m and c from BET, n_m and K from Langmuir, Mn and Mw by hand, quantum yields summing to 1
SCI-46	Numerical chemistry problems, from basic to expert, must give textbook values and balance reactions by themselves	`bsi-chi-esame.js` — 40 models on four levels (General and Inorganic, Analytical, Physical Chemistry, statistical thermodynamics and activities), IUPAC atomic masses, reactions balanced via the composition-matrix null space, exact pH from the charge balance	`test_esami` — $Ca_3(PO_4)_2$, glucose from composition, limiting reagent, ICE, acetic acid, 10^{-8} M HCl at pH 6.98 (**both ways**: not 8), AgCl and Ag_2CrO_4, Daniell, Faraday, Hα, Hess, titration, H_3PO_4, $BaSO_4$, HCl B, argon Sackur-Tetrode, Pourbaix, ionic strength; every solved paper
SCI-47	Astrophysics, nuclear physics, quantum physics and PhD-level problems must reproduce independent known values	`bsi-fis-esame.js` (astrophysics, nuclear, quantum, variational, perturbation), `bsi-chi-esame.js` (Marcus, isotope effect, Wigner, organic, spectroscopy, biochemistry), `bsi-mat-risolutore.js` (residues, heat equation), `bsi-percorsi.js`	`test_esami` — Sirius, Proxima, Kepler for Earth and Jupiter, Schwarzschild, ^{56}Fe and ^{235}U, D-T, ^{131}I, ^{14}C, 2p orbital, Larmor at 9.4 T, variational helium, perturbation = exact, Marcus, KIE, aspirin DBE, lysine pI, Cl_2 pattern, $\int 1/(x^4+1) = \pi/\sqrt{2}$, heat equation against the exact solution
SCI-48	An assistant reading natural-language questions must say what it understood and where each datum comes from, and the result must be the one computed another way	`bsi-assistente.js` — rules for mathematics (limits, derivatives, integrals, ODEs brought to the form G(x, y, y′, y″) = 0, series, matrices), IDF keyword scoring and re-ranking by data used for physics and chemistry, unit conversion, inverse problem by bisection; `bsi-mat-cas.js` (trigonometric integrals, e^(c·ln u), polynomial GCD)	`test_assistente` — 80 questions: model understood, data read (0.1 M is not 0.1 m; «carbon 14» is not a datum; «brakes», «falls», «an electron»), result against the independent formula; **both ways**: an arbitrary sentence does not produce an invented model; Spectra's tools with a question and with an explicit model
SCI-49	What is shown must be right: an animated or three-dimensional figure must not show a shape theory rules out, and an animation must end exactly on the picture the module would have drawn	`bsi-lab3d.js` — hydrogen-like orbitals sampled exactly from	ψ

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SCI-50	A simulation must recover the exact answer where one exists, and declare its approximations elsewhere	`bsi-fis-meccanica.js` (projectile with quadratic drag by RK4, collisions with restitution, pendulum with the elliptic integral, normal modes), `bsi-fis-onde.js` (conjugate points, N-slit diffraction), `bsi-fis-circuiti.js` (closed-form RLC transients, impedance, Bode), `bsi-fis-termo.js` (cycles with per-process balances, hard-disc gas with elastic collisions, random walks), `bsi-mat-numerici.js`, `bsi-mat-geometria.js`, `bsi-chi-equilibrio.js` (mass-action kinetics against the computed equilibrium, van 't Hoff with ΔH° and ΔS° declared constant)	`test_laboratori` — range and flight time in vacuum, $T/T_0 = 1.18034$ at 90°, energy lost $\frac{1}{2}\mu(1 - e^2)v_{rel}^2$, Kirchhoff in the three RLC regimes, −3.01 dB and −45° at cut-off, ΔU = ΔS = 0 around cycles and efficiencies equal to the formulas, Maxwell-Boltzmann mean speed and hard-disc pressure, numerical method orders, Runge's phenomenon, conic foci, skew lines, kinetic equilibrium = computed equilibrium, Le Chatelier **both ways** (exothermic/endothermic) and in the page. **Found and fixed**: with a low hot temperature and high r the Otto cycle «burned» by cooling (efficiency −130 %)
SCI-51	Astrophysics, relativity, fluids, vector calculus and complex kinetics must reproduce known values and theorems	`bsi-fis-astro.js`, `bsi-fis-relativita.js`, `bsi-fis-fluidi.js`, `bsi-mat-vettoriale.js` (symbolic curl and divergence with a numeric zero check), `bsi-chi-cinetica.js`	`test_scienze` — ΛCDM age 13.8 Gyr and Einstein-de Sitter $2/(3H_0)$, integrated Planck = σT^4, Earth-Mars Hohmann 259 days, Earth transit 84 ppm, HD 209458 b K, Kepler on the planets; twins and invariant interval; Poiseuille R^4, Stokes, Jurin; Green in both forms, the vortex with zero curl and circulation 2π **both ways** (with and without the origin inside); consecutive t_max, Hopf threshold and Brusselator period
SCI-52	Probability, visual mathematics, electromagnetism, sound, semiconductors, distillation and voltammetry must reproduce known results, and models must declare where they deviate from measurement	`bsi-mat-probabilita.js`, `bsi-mat-visiva.js`, `bsi-fis-em.js`, `bsi-fis-suono.js`, `bsi-fis-solido.js` (Varshni, N_c and N_v ∝ T^(3/2)), `bsi-chi-distillazione.js` (Antoine, van Laar, McCabe-Thiele), `bsi-chi-voltammetria.js` (finite differences with a second-order implicit boundary, Butler-Volmer)	`test_esplorazioni` — Bayes 1/6, central limit $\sigma/\sqrt{n}$, exact Fourier, logistic map periods, closed-form dipole flux = integral, n_i of Si/Ge/GaAs, V_bi, Antoine boiling points, ethanol-water azeotrope against measurement, McCabe against Fenske, reversible peak = Randles-Ševčík within 1 %, ΔE_p 58 mV, irreversible peak = 0.4958 $nFAC\sqrt{(\alpha nFvD/RT)}$. **Found and fixed**: with a first-order boundary gradient the simulated peak was 95.6 % of theory and ΔE_p 63.5 mV

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SCI-53	The nineteen new sections (numbers and cryptography, graphs and optimisation, sequences and series, calculus of variations, topology, algebraic structures, game theory, logic and computability, statistical inference, non-Euclidean geometries; gravitation, heat, Ising, particles, analytical mechanics, lasers, climate; phase diagrams, potential energy surface) must reproduce known results or an independent second method	`bsi-mat-numeri.js`, `bsi-mat-grafi.js`, `bsi-mat-serie.js`, `bsi-mat-variazioni.js`, `bsi-mat-topologia.js`, `bsi-mat-algebra.js`, `bsi-mat-giochi.js`, `bsi-mat-logica.js`, `bsi-mat-inferenza.js`, `bsi-mat-noneuclidea.js`, `bsi-fis-gravita.js`, `bsi-fis-calore.js`, `bsi-fis-ising.js`, `bsi-fis-particelle.js`, `bsi-fis-analitica.js`, `bsi-fis-laser.js`, `bsi-fis-clima.js`, `bsi-chi-fasi.js` (IAPWS, Span–Wagner), `bsi-chi-pes.js` (Sato LEPS)	`test_approfondimenti` — π(10⁵) = 9592, RSA d = 2753, Dijkstra = Floyd–Warshall, Kruskal = Prim, figure-eight, Earth–Moon Lagrange points, Onsager, boiling at 373.124 K, H_3 saddle at 0.93 Å and 0.42 eV. `test_pareggio` — 462 checks in twelve scripts in `verifiche/`: π²/6, the cycloid faster than any other curve, χ of the solids from their vertices, 549 946 tic-tac-toe nodes, busy beavers, table t quantiles, Girard, pion momentum 29.79 MeV/c, exact pendulum period, Rabi against Schrödinger, 255 K and 288 K. **Found and fixed**: the first LEPS had its asymptote at −6.40 eV instead of −D_e; the hyperbolic tiling side length had p and q swapped; the Noether demo potential let the particle escape
SCI-54	The Articles section must have the **same level** for chemistry, physics and mathematics: same tabs, same rules, no year or name written by hand	`bsi-articoli.js` (three subjects), `bsi-articoli-nobel.js` (135 explanations), `bsi-articoli-nobel-fisica.js` (140), `bsi-articoli-premi-matematica.js` (140; Abel Prize and Fields Medal from Wikidata)	`test_articoli` — for each of the three subjects at least 130 explanations, none under 450 characters, no year, no duplicate expression, eight article searches; fake answers injected for the Physics Nobel and for Wikidata: the names received appear and no others
UI-18	Each subject must have its own bar: whoever studies physics or mathematics must not scroll through the chemistry groups	`bsi-materie.js` — two round buttons above Spectra and ✨, always the two subjects you are not in (flask, atom, sigma); the bar shows the subject's groups plus Study and Utility; opening a section of another subject switches subject by itself; the choice is remembered	`test_approfondimenti` — visible groups for each subject, two different icons, accessible name, position above Spectra, subject remembered after reload, no section unreachable

ID	Requirement	Implementation	Verification bench
SCI-55	The ten sections of the Astrophysics group (solar system, orbital mechanics, celestial mathematics, stars, black holes, gravitational waves, galaxies, cosmology, planetarium, exoplanets) must reproduce published values or an independent second method, and declare illustrative data	`bsi-astro-sistemasolare.js` (Standish JPL elements, Meeus), `bsi-astro-orbite.js`, `bsi-astro-matcelesti.js`, `bsi-astro-stelle.js` (Lane–Emden, AME2020), `bsi-astro-buchineri.js` (Schwarzschild), `bsi-astro-ondegrav.js`, `bsi-astro-galassie.js` (Freeman, NFW), `bsi-astro-cosmo.js` (Planck 2018), `bsi-astro-cielo.js`, `bsi-astro-esopianeti.js`	`test_v237` — ten scripts in `verifiche/`, 314 checks: Meeus Julian date 2436116.31, Curtis ex. 4.3, elements ↔ state over 2000 orbits ($2.8 \cdot 10^{-14}$), Laplace limit 0.6627434, $\xi_1 = 6.89685$ for n = 3, Mercury's perihelion 42.982″, deflection 1.7512″, GW150914 chirp mass 28.096 $M_\odot$, Freeman peak at 2.150 R_d, ΛCDM age 13.791 Gyr, GMST of Meeus ex. 12.a, Earth's T_eq 254.8 K
SCI-56	The ten new physics laboratories (spin and qubits, statistical mechanics, nuclei, materials, electronics) and mathematics laboratories (PDEs, complex analysis, Markov, SVD, codes) must reproduce known results or a second method	`bsi-fis-spin.js`, `bsi-fis-statistica.js`, `bsi-fis-nuclei.js` (AME masses), `bsi-fis-materiali.js`, `bsi-fis-elettronica.js`, `bsi-mat-edp.js`, `bsi-mat-analisicomplessa.js`, `bsi-mat-markov.js`, `bsi-mat-svd.js`, `bsi-mat-codici.js`	`test_v237` — ten scripts in `verifiche/`, 391 checks: HZH = X, Schottky peak (0.41678 Δ; 0.43923 k), B/A(^{56}Fe) 8.7903 MeV, Q of D + T 17.589 MeV, I-section moment with Huygens–Steiner, diode with Newton = Lambert W, Bessel zeros ($j_{01} = 2.404825557695773$), Cauchy–Riemann on nine functions, stationary distribution by powers = Cramer, σ of [[1,1],[0,1]] = φ and 1/φ, CRC-32 = 0xCBF43926, 112 single errors corrected by Hamming(7,4)
SCI-57	Study material must be checked: minimum quantities, valid schema, quiz answers computed where possible, tables with a numeric proof, citations in the declared formats	`bsi-studio.js` (`valida`, `MINIMI`), `bsi-studio-fisica.js`, `bsi-studio-matematica.js`, `bsi-risorse-materie.js` (citations, tables with `prova`)	`test_v237` — `verifica-studio` (167 checks on the engine: generator, shuffle, Leitner, score, HTML cleaning), `verifica-studio-fisica` and `verifica-studio-matematica` (minimums and schema; **264 mathematics answers recomputed** out of 346 questions), `verifica-risorse` (87 checks: APA 7, IEEE, ISO 690, BibTeX on hand-built records, table proofs)

ID	Requirement	Implementation	Verification bench
UI-19	Every tab of the twenty new sections must open full, without warnings, with its canvases drawn and an accessible name, on a large screen and on a phone, without blocking the page	group 9 «🔭 Astrophysics» in `index.html`; sections built with `bsi-fisica-comune.js` and, for 3D, `bsi-lab3d.js`	`test_v237` — at 1280 and 390 px every tab is full, has no ⚠, every canvas draws (for 3D the scene is sampled) and has a name; **2D tabs within 250 ms, 3D tabs within 6 s** (in the bench WebGL runs in SwiftShader, with no GPU); the Astrophysics group has at least 12 sections; nothing overflows at 390 px; no JavaScript errors; the section names in English
UI-20	«Check for updates» must always give an answer: up to date, new version, site unreachable	`bsi-aggiorna.js` (service worker version via `BSI_VERSIONE`, `sw.js` from the same site with no cache); in `index.html` the 🔄 Updates window, with GitHub as fallback; in `sw.js` the answer to `BSI_VERSIONE`. **Found and fixed**: the window looked for a `CACHE_VERSION` line that `sw.js` no longer has, and on GitHub: the published version was never read	`test_controllaagg` — 12 checks: the «Check» button actually pressed in the three cases (same `sw.js`; `sw.js` with a higher version (v999); request aborted), «Update now» only in the second, and `BSIAggiorna.controlla()` with its pop-up notice in the same three cases
SCI-58	The 3D tabs added to 29 chemistry, physics and mathematics sections must show quantities computed from the section's model and compared with known values, a closed form or a second method	`bsi-chi-pes.js`, `bsi-chi-fasi.js`, `bsi-chi-cinetica.js`, `bsi-chi-equilibrio.js`, `bsi-fis-cristalli.js`, `bsi-fis-schrodinger.js`, `bsi-fis-hartreefock.js`, `bsi-fis-rotovib.js`, `bsi-fis-dinamica.js`, `bsi-fis-meccanica.js`, `bsi-fis-em.js`, `bsi-fis-onde.js`, `bsi-fis-gravita.js`, `bsi-fis-relativita.js`, `bsi-fis-fluidi.js`, `bsi-fis-termo.js`, `bsi-fis-solido.js`, `bsi-fis-ising.js`, `bsi-fis-particelle.js`, `bsi-mat-multi.js`, `bsi-mat-vettoriale.js`, `bsi-mat-calcolo.js`, `bsi-mat-lineare.js`, `bsi-mat-ode.js`, `bsi-mat-probabilita.js`, `bsi-mat-topologia.js`, `bsi-mat-noneuclidea.js`, `bsi-mat-complessi.js`, `bsi-mat-variazioni.js` (scene through `bsi-lab3d.js`). **Found and fixed before publication**: `applicaMobius` did not recognise the pole for an angle of exactly π; in molecular collisions the normal was taken from spheres that had already overlapped (fraction above threshold 0.345 against e^{-1} = 0.368, now 0.370)	`test_3d238` — runs the 29 scripts `tools/banchi/verifiche/verifica-<id>-3d.js` (for `smattopo` and `smatnoneucl` the section's own script): 311 checks, among them the LEPS saddle with a single imaginary frequency, h*/R = 1.32549 for the soap film, for 3D Ising

ID	Requirement	Implementation	Verification bench
SCI-59	The Spectra Studio lessons must show numbers computed from a model and compared with the exact solution, an experimental value or a second method	`bsi-spectra-studio-lezioni.js` — `calcolo.caduta` (semi-implicit Euler, optional quadratic drag), `calcolo.vsepr` (minimum of the energy of domains on the sphere, lone pairs with w = 1.25), `parziale`, `coda`, `fourierX2`, `coprimi`	`test_spectra_studio` — runs `verifiche/verifica-studio-spectra.js` (23 checks): fall time √(2h/g) = 2.0193 s for h = 20 m and the two spheres together; range 12.116 m; with air the thrown one lands later; tetrahedron arccos(−1/3) = 109.471°, trigonal 120°, linear 180°; water 104.40° (experimental 104.5°, w calibrated); ammonia 106.69° with the same w (prediction); S(100 000) within the tail of π²/6; tail between 1/(N+1) and 1/N for N = 10, 100, 1000; Fourier series of x² at π = π²; harmonic series as ln N + γ; coprimes → 6/π² within 3σ; each lesson with exactly one right answer, every option explained, at least three proof steps, exercises on four levels
UI-23	Every tab of the 29 sections with the new 3D tab must open full, without warnings, with its canvases drawn and an accessible name, and the last tab must be a 3D scene that draws; a chart without a name must take the tab's	`bsi-fisica-comune.js` — `prepara()` and `mostra()` give canvases without a name (or with a one-letter name) «Chart: <tab> — <section>»; the 3D scene is built after the click, outside the handler	`test_3d238` — at 1280 and 390 px, for every section: all tabs full, no ⚠, every canvas draws and has a name of at least three characters; the last tab is a scene with `Scena.campiona() > 0`; **synchronous work on click under 250 ms** (WebGL in SwiftShader); nothing overflows at 390 px; no JavaScript error
UI-22	The Spectra Live console must stay usable on a phone with Android's status bar and navigation keys over the page, with no overlapping elements, and every section must be reachable from it	`bsi-spectra-live.js` — `env(safe-area-inset-*)` margins with minimums of 34 and 46 px (class `.bsiJV-bordi`) when the window covers the screen on a touch device or the app is installed; layout computed in the drawing loop; 🧭 navigator per subject; context shortcuts; 💬 button; round Spectra Live button (since v239 Spectra's only button, UI-25)	`test_spectra238` — on a simulated phone (412×915, touch): safe margins active, first button ≥ 34 px from the top edge, controls ≥ 46 px from the bottom edge, telemetry in two columns not covering the face; on a large screen no forced margins; on both, header, face, state, dialogue, controls and attribution without overlaps; the navigator lists, for each of the three subjects, exactly the sections of the bar, the quick tools and the phrases; opening a section of another subject switches subject and leaves the Spectra Live button

ID	Requirement	Implementation	Verification bench
UI-24	Every section must have a guided lesson built from the section itself: one chapter per tab, the controls shown at work and put back as they were, a check question with its explanation, a summary; during the lesson no floating button over the panel, and at the end everything as it was	`bsi-lezione.js` — `BSILezione.costruisci()` reads title, subtitle, tabs, explanations, canvases, sliders and selects; check question from the subject's study kit (most words in common with the section); player `#bsiLZ` with spotlight, ◀ ▶, pause and voice; «🎓 Guided lesson with Spectra» button in every section; demonstrated controls return to their starting value even when skipping ahead or closing; floating buttons hidden and restored; «reduce motion» respected. `bsi-spectra-live.js` — «give me a lesson» («fammi lezione») command and 🎓 shortcut	`test_lezione` — 30 checks: going through all **194** sections of the bar in the three subjects, each has its lesson (opening … summary) and the button, and each of the **99** with tabs has one chapter per tab; at least 60 lessons with a check question (**79** at the time of writing); at 390 and 1280 px: the lesson starts from the opening, floating buttons are hidden, a slider really changes value under the spotlight and returns to its starting value when skipping ahead half-way, a select goes through its options and returns, the check shows the right answer and the explanation, the summary lists the chapters, nothing overflows, on closing the panel is gone and the buttons are visible again; «fammi lezione» starts the lesson; no JavaScript errors. The voice is not tested by the bench
UI-25	Spectra must have a single visible button, its face in a circle, from which both console and chat are reached	`bsi-spectra-live.js` — `#bsiJV-fab` (58 px, round) in place of the `#bsi-spectra-fab` pill, which stays in the document hidden by the `bsiJV-solo-volto` class because other modules and the benches use it; the circle opens the console, 💬 opens the chat; vector face at once, portrait after `load`	`test_spectra238` — after loading the circle is visible, the pill is not, and the circle has `border-radius: 50%`

ID	Requirement	Implementation	Verification bench
UI-26	Spectra Studio must lead every lesson through the four phases predict → test → prove → reveal on a large screen and on a phone, remember the predictions, recommend the exercise level and remain usable without WebGL	`spectra-studio.html` — three.js scene with Spectra from the shared controller of `bsi-spectra-figura.js` (embedded body `models/spectra-corpo.bsi` since v241, rigged, and «Noir» face on the collar; while it arrives the `#invito` notice «Spectra sta arrivando…», if it does not arrive «↻ Riprova»; no substitute, no «🧍 Corpo» button); panel `#pan` with all the text, `aria-live`, lesson tabs with `role=tab`; memory `bsi-spectra-studio-v1`; voice switch 🔊/🔇 remembered in `bsi-spectra-studio-voce`; without WebGL no scene, panel and measurements remain. `bsi-spectra-live.js` — «Spectra Studio» entry in the 🧭 navigator	`test_spectra_studio` — at 1280 and 390 px the three lessons go through the four phases with the measurements in the scene (fall, water, Basel: S_{1000} = 1.64393, coprimes 0.6079), the real body is always there without loading it and the notice is gone, no «🧍 Corpo» button and no file choice, the published file is a «BSIC» block and not a `.glb`, an invalid block rejected without errors, the face rests on the real body's collar; for every lesson four predictions, measurements in the table and formulas in the proof, the reveal marks the right answer with four exercises of which one recommended, the memory remembers the prediction and the exercise opened; at 390 px nothing overflows; without WebGL the lesson gives the measurements (2.019 s); no JavaScript errors

ID	Requirement	Implementation	Verification bench
UI-27	In every AI section (Spectra AI and the three areas with the assistant) Spectra must appear with the body chosen by the author, **always present without having to load it** (since v241), under the title, and react to what the app does; the icon must be the face only; no substitute; the sections outside `#main` must have no empty space above them	`bsi-spectra-figura.js` — `Spectra()` (`carica()`: the embedded body `models/spectra-corpo.bsi`, reopened by `apri`; an old body in IndexedDB `bsi-spectra-modelli` is deleted; in the app, with `soloDispositivo`, no requests for the development files, no 404s; «Spectra sta arrivando…» and «↻ Riprova»), `monta()` (lab-style frame, «SPECTRA · IN ASCOLTO / PENSA / PARLA», «🎙️ Parla con me», «🧠 Spiega», «🎓 Studio»), `sezioni()` (one figure moved into the open section); events `bsi-spectra-stato`, `bsi-spectra-risposta`, `bsi-spectra-dice` (from `bsi-spectra-live.js` and `bsi-assistente.js`); `index.html` — class `bsi-sez-fuori`	`test_spectra_figura` — 103 checks at 412 and 1280 px (28 of them for «🧠 Spiega», AI-18, 14 for the hologram and 10 for the catalogue, UI-28): button face only, pill hidden; in the four sections the figure inside the section under the title (≥ 280 px) and only one in the whole app; `#main` at 0 px above the sections outside it; Spectra's body is always there, without loading it (embedded, bones 20/23); no «🧍 Corpo» button and no file choice; on the real body face on the collar (0.0 cm), only the face visible; thinks, explains with lip sync on body and face, doubt on error, speaks with Spectra Live, explains the local assistant's working; on the phone the buttons do not cover the floating ones; no JavaScript errors. `test_spectra_studio` — in the studio the embedded body is always there and the published file is a packed block. **Not tested**: the visual rendering of the body, which has to be judged by looking

ID	Requirement	Implementation	Verification bench
UI-28	To Spectra's right a hologram must show live scenes of the sections (computed 2D and 3D animations, or the open section live), follow the conversation's subject and lead to the section with one tap; **every section** of chemistry, physics and mathematics must be showable and explainable there («non deve mancare nulla in questo ologramma»), including those without graphs; and **Spectra's figure must not disappear**: showing a section must not take the user away from the AI section, and closing the window must return to where the figure is	`bsi-spectra-figura.js` — `Ologramma` (glass screen with base and beam, on the viewer's right), `SCENE_OLO`: planets (Kepler's equation M = E − e sin E solved by Newton every frame, T ∝ a^1.5), molecule (benzene with the π cloud), spectrum (IR with Gaussian bands and a cursor), surface (z = sin(r − t)/r), orbital (3d_z² with phases), wave (Gaussian packet with free Schrödinger dispersion), screen (the section behind the window, re-rendered with `BSISpectraLive.cattura` 5 times a second); `sceneDi(text)` picks the scene from the words; event `bsi-spectra-tema` from `bsi-spectra-spiega.js`; «… · apri ›» label that opens the window on «🧠 Spiega» and the section behind it (`BSISpiega.mostraSezione`), without leaving the AI section; on closing the window the return to the starting AI section (`sai`, `sareachim`, `sareafis`, `sareamat`) and the tour stopped; «indica» gesture before «spiega»; button bar under the scene; in «schermo» a section without graphs is projected as a card (title, tab, up to 6 controls with values, visible text). `bsi-spectra-spiega.js` — `catalogo()` (every `.nav-group-panel [data-s]` button of the bar with group and subject from `BSIMaterie.materiaDi`, not written by hand), the «🗂️ Tutte le sezioni nell'ologramma» card with search and subject filter, `mostraSezione(id)` (`BSISpectraLive.calcolo.apriSezione`), `giro(materia)` / `giroFerma()` («🎞️ Mostrami tutto», one section every 7 s), `sezioneDi(testo)` (the section the question names: words with stems, weighted acronyms, synonyms)	`test_spectra_figura` — 12 checks per width (412 and 1280 px): the catalogue has every section of the bar (194/194: chemistry 75, physics 60, mathematics 30, shared 29); a question that names a topic finds its section (titolazioni → `stitr`, Schrödinger → `sschrod`, integrali per parti → `smatcalc`, spettro di massa → `sms`, spettro infrarosso → `sirvis`, meccanica orbitale → `sastroorbite`, «ciao come stai» → none); asking about titrations Spectra opens the section behind the window, shows it in the hologram and explains it with its data; the Periodic Table (no graphs) is projected as a card; «🎞️ Mostrami tutto» runs through the subject's sections; the hologram is to Spectra's right (screen x); six live scenes; perihelion/aphelion speed ratio = (1+e)/(1−e) = 1.94 within 8 %; the subject follows the conversation (IR answer → spectrum, orbit question → planets); the label leads to the section; a tap on the label opens «Sistema solare 3D» **behind** the «🧠 Spiega» window: Spectra stays visible in front and the hologram shows the section live; once the window is closed, the app returns to Spectra's section and the figure is there; also, with the 3D scene open behind «🧠 Spiega», the hologram shows it live, and on answering Spectra points at the hologram and then explains with the hands. **Not tested**: the visual rendering of the scenes, which has to be judged by looking

ID	Requirement	Implementation	Verification bench
UI-14	One function must not have two entry points. The spectra lived in **nine sections** across two menus; Spectra in three places; and the ✨ panel showed **two "Data Science", two "File Manager", two "Articles", two "Updates"**, because the panel has a hardwired list that opened the very sections the menu module was inserting — and nobody compared the two lists	`bsi-centrospettri.js` — the panels are **moved** (`appendChild` moves, it does not copy) into a single tabbed section; the old buttons stay hidden because they are the target of `goSection()` , and a tab **clicks the button** instead of showing the panel, so lazy initialisation goes through the one road that already existed. `bsi-menu.js` — the "Utility" category is in the **main bar** with all its entries, and the rule is: if something has a **reachable** button in the bar it is not in the ✨ menu, if it has none it is only in the menu. "Reachable" is not measured with `offsetParent` — the bar shows one group at a time, and a perfectly reachable button has a null `offsetParent`	`test_articoli` §5-6 — the **ten** tabs are inside the centre, and each identifier exists in the document **exactly once** (the proof that it was moved and not copied); `goSection()` on each opens the centre on the right tab and shows **only** one. `test_menu` — Utility visible in the bar with all **fifteen** entries plus the two language ones, **no** menu entry repeating a bar section, **no** duplicate label in the panel, across **two viewports**: large screen and phone. **Both ways**: no repeated id, no duplicate button, Spectra and the reader out of the bar but with their button still in the document because `goSection()` uses it
UI-17	Anything that moves must be pausable, slowable and restartable, must not consume when not visible, and must honour «reduce motion» (WCAG 2.2.2)	`bsi-fisica-comune.js` — `F.anima` : bar with ⏸/▶, ↺, speed ×¼…×4 and time; step with real dt capped at 1/20 s; stops off-screen (IntersectionObserver), with the page hidden and when the canvas is removed; with «reduce motion» it starts paused with the first frame drawn. Every laboratory uses it	`test_scienze` — time advances, stops when paused, runs faster at ×4 than at ×¼, stops off-screen and resumes when visible; with «reduce motion» it starts paused and ▶ starts it; every bar has accessible names
UI-15	A data table must not exist twice: two copies diverge, and the reader gets a different answer depending on where they came in	The J coupling constants existed in two independent tables with **different** numbers (aromatic ortho ^{3}J 6–9 Hz against 7–10 Hz). The one inside the Spectra centre remains, which sits in the section with its own reference list; `_apriCostantiJ()` is the single road and leads there	`test_articoli` §6 — asking for the J constants opens the centre on the reference-tables tab, on the right sub-tab, and the table has its rows

ID	Requirement	Implementation	Verification bench
ING-02	A hand-written list of articles is stale the next day, and a prize year written from memory is an error nobody can see	`bsi-articoli.js` reads **live** from Crossref and from the Nobel Foundation; `bsi-articoli-nobel.js` holds **explanations only**, matched to the topic of the official motivation and not to the year, so no attribution comes from the application. The addresses are not written: they are **pure functions** of the identifier (DOI, year)	`test_articoli` §1-4 — the links tested as functions (stable for the same datum, different for a different one); the explanations matched against nine test motivations and **none** invented when the topic is not recognised; and the proof that counts: a response is **injected** with a name that exists nowhere, and the screen must show **that** name. **Three distinct failures tested**: failed request, empty answer, unrecognised shape — and they must be stated **differently**
UI-16	Changing language must cost a gesture, not a navigation	`bsi-sellingua.js` — a **round** button top right, at the title's height, with the flag in use; it opens into a disc with the fourteen flags on **two shells** (six inside, like a benzene ring) and, at the centre, the language name and **how much it is translated, measured** by `coperturaPerLingua()`. The actual switch stays in `bsi-lingue.js`: here there is only the way to call it, and the "Language" section remains because it INFORMS while this one ACTS	`test_sellingua` across **two viewports** — round, top, right, and "at the title's height" **measured** as the overlap of the two boxes (with `align-self:center` it lined up on a large screen and not on a phone, because there the flex aligns to the top and the logo block is two lines tall); an accessible name that says what it does rather than "flag of Italy"; the disc **inside the screen**; and **both ways**: the language really changes and going back to Italian the text is identical
ING-01	One function must not have two implementations: the React/FastAPI stack must run THE SAME files as the application	`stack/api/worker/motore.mjs` — a Node process loads the repository modules with RDKit WebAssembly; no Python engine	`test_stack` — **15 checks**: compares the SHA-256 fingerprints declared by `GET /salute` with those of the files on disk, verifies the numbers match the browser's, and that no `.ts` / `.tsx` file contains a shift table

2. AI agent requirements

ID	Requirement	Implementation	Verification bench
AI-01	A model retired by the provider must not block the assistant	`modelloSuggeritoDaErrore()`, `bsi_modelli_ko` blacklist	`test_404` — 36 checks

ID	Requirement	Implementation	Verification bench
AI-02	A provider unreachable from the browser must be remembered and demoted	`bsi_prov_ko`, 24 h TTL, `providerUtilizzabili()`	`test_ko` (35), `browser_ko` (37)
AI-03	A temporary overload must produce a retry, not an error	`erroreTemporaneo()`, `TENTATIVI_TEMPORANEO`	`test_503` — 40 checks
AI-04	An imposed wait longer than 10 s must switch provider if a ready one exists	`ATTESA_TROPPO_LUNGA_MS`	`test_attesalunga` — 14 checks
AI-05	The provider's token ceiling must be learned from the error, not hard-coded	`tettoDaErrore()`, `MARGINE_TETTO`	`test_tetto` — 32 checks
AI-06	Opaque reasoning signatures must be returned to the provider	capture and replay in `appendAgentTurn()`	`test_firma` — 31 checks
AI-07	Two overlapping submissions must not corrupt the history	`_turnoInCorso` in `send()`	`test_doppioinvio` — 7 checks
AI-08	The user must be able to measure which providers they can actually reach	`provaTuttiIFornitori()`	`browser_prova` — 29 checks
AI-09	Paid providers must not be offered as free ones	separation by `PROVIDERS[id].free`	`browser_prova` §D2
AI-10	A documented route must exist to remove CORS and rate limits	`proxy/spectra-proxy.js` + in-app guide	`test_guidaproxy` (12), `browser_proxy` (15), `browser_proxyui` (17)

ID	Requirement	Implementation	Verification bench
AI-11	Spectra's face must really move with what he says and hears: emotions recognised in sentences, a face that changes, eyes in the theme's colour	`bsi-spectra-androide.js` — «Noir» model (Skyfox13, CC BY 4.0, modified); 17 emotions from 11 expressive fields blended on the CPU into one attribute (WebGL allows at most 16 per vertex); mouth from the spoken text; fallbacks `bsi-spectra-robot.js` and `bsi-spectra-volto.js`	`test_v237` — 17 emotions and 11 fields; nine sentences give the expected emotion; eyes at rest `#1a3050`; more than 1000 animated vertices; face still in neutral and different between joy and empathy; the set emotion persists; a compliment produces the warm smile; more than 1000 pixels drawn at the centre; no JavaScript errors. The no-WebGL fallback is not exercised by the bench
AI-12	Spectra's mind must stay in the browser and declare itself for what it is: a state that responds to events, not consciousness	`bsi-spectra-mente.js` — key `bsi-spectra-mente-v1`, state of curiosity, attention, care, mood, bond; `coscienza()` passes a summary to the model; «forget everything»	`test_v237` — a compliment raises the mood, «how are you» answers from the internal state, the summary for the model contains the mood. «Forget everything» and the persistence of the key are not exercised by the bench
AI-14	The face must show how demanding the question is and stay alive without looking conscious: concentration from the section's level and from the question, eyelids and eye tint per emotion, sleep when nobody is there	`bsi-spectra-live.js` — `difficolta(testo)` and `livelloSezione(id)` (first level of the path in `bsi-percorsi.js`); `bsi-spectra-androide.js` 3.0 — eyelids as a deformation of each eye's vertices around its centre, tint per emotion in the `#1a3050` family, `concentra()`, damped springs, micro-expressions, sleep after 120 s without contact	`test_spectra238` — an advanced-course question weighs more than a sum, «in simple words» lowers it, the same question weighs more in a PhD-level section than in a school-level one and the level is the path's; joy narrows the eyes, surprise widens them, doubt makes them asymmetric, `#1a3050` at rest, four emotions give four tints, concentration narrows and brightens the eyes, the blink closes the eyelids, after two minutes it sleeps and wakes at the first touch

ID	Requirement	Implementation	Verification bench
AI-15	Spectra's body must work with any humanoid avatar: recognise bones, blendshapes and hair, report what is missing, and drive gaze, gestures, emotions, lip sync, eyelids, walking and sitting; the hair must turn black	`bsi-spectra-corpo.js` — recognition by name (Mixamo, Ready Player Me, VRM, Unreal, generic), `rapporto()`, `orienta()` (poses as world directions), emotions as ARKit weights, `capelliNeri()` (colour `#0a0a0c` and texture desaturated in the shader), `fai`, `cammina`, `siedi`, `audio`; test `manichino()`	`test_spectra_studio` — on the mannequin (11 checks): bones 22/23 (no jaw), blendshapes 14/14 and hair found; red hair turned black (RGB 0.039 0.039 0.047); arms down from the T-pose; pointing with the forearm on the target (1.7° off); thinking with the hand near the chin; the gaze turns; walking moves the body 0.96 m moving the legs; sitting lowers the hips by 42 cm; joy → smile, worry → mouth corners down; lip sync from audio opens the jaw; the blink closes the eyelids. The fallback to the «cyberpunk» model (`models/spectra-cyberpunk.glb`) was removed in v241: the body is the embedded one (AI-17)
AI-16	The quality voice must go through a measurable effects chain, give the lip sync from the audio and fall back on the browser's synthesis without a key	`bsi-spectra-voce.js` — `catena()` (bass, mud, presence, ring, comb, saturation, reverb from `impulso()`, compressor), `leggiBocca()` (energy and centroid), `qualita()`, `TIMBRO`; `bsi-spectra-live.js` and `bsi-lezione.js` use it when there is an OpenAI key	`test_spectra_studio` — in an offline audio context (3 checks): at 80 Hz the bass comes out boosted (6.9 dB), the reverb leaves a tail after the sound ends, without a key `qualita()` is `'sistema'`. **Not tested**: the call to OpenAI and listening to the voice

ID	Requirement	Implementation	Verification bench
AI-17	A humanoid body without a skeleton must become drivable by the body driver without external tools: a recognised skeleton, weights that do not tear the torso, a hood rigid with the head with an opening for the «Noir» face, the app's colours, a definition suited to high-density screens	`bsi-spectra-rig.js` — `misura()` (height, axis, legs, hands, shoulders; human canon for the points that cannot be measured), skeleton with Mixamo names (20 weighted bones and 7 tips), weights ∝ 1/d⁴ with a per-region gate, normal, geodesic-distance-from-the-hand and head-axis filters, three smoothing passes; `rigga()` with `cappuccio`, `gonfia`, `viso` (elliptical opening in the shader), `suddividi` (Phong tessellation) and the options `visiera: false` (helmet as matte shadow) and `testa` (scale), which the studio does not use; bump map from the texture's luminance; `ricolora()` (night blue, turquoise → emissive `#00c9b7`). `spectra-studio.html` and the figure use it on the **embedded** body `models/spectra-corpo.bsi` (since v241; packed by `tools/impacchetta-corpo.js`, reopened by `apri` in `bsi-spectra-figura.js`); only in the studio, in development, if the block is missing, `models/spectra-corpo.glb` (glossy black visor kept, pixel ratio up to 3); `bsi-spectra-androide.js` — «solidale» face	`test_spectra_rig` — 30 checks on a synthetic A-pose humanoid: 27 bones with Mixamo names recognised by the driver, locomotion; weights summing to 1, the hand to the hand and forearm, the flank next to the hanging hand with zero arm weight, legs never crossing; in the «spiega» gesture the hand moves 0.52 m and the flank 1 mm; walking moves the body; with the face only no triangle removed, ellipse in the shader, double side, hood rigid with the head (480/480 vertices), with `visiera: false` the helmet in shadow without removing vertices; night blue and teal light, grey emits nothing; tessellation ×4 with weights normalised on the new vertices and the skin closed (open border edges only double); no JavaScript errors. `test_spectra_studio`, on the **real embedded body**: face–hood sync in 24 poses (threshold 1 mm, measured 0.0 mm), of the «Noir» model only mask, eyes and details visible, «solidale» face; at 1280 and 390 px the body is always there without loading it, no «🧍 Corpo» button and no file choice, the published file is a «BSIC» block and not a reusable `.glb`, a block that is not a model rejected without errors. **Not tested**: the visual rendering of the folds and the hood on the real body, which has to be judged by looking

ID	Requirement	Implementation	Verification bench
AI-18	«🧠 Spiega»: exhaustive explanation of chemistry, physics and mathematics from high school to advanced PhD, including of the screen (spectra, graphs, 3D scenes)	`bsi-spectra-spiega.js` — `BSISpiega.sistema(level, subject)` (five levels with `PROFONDITA`, «do not simplify below this level and do not leave out anything relevant», nine-part mandatory structure with the quantitative reading of NMR/IR/MS/UV-Vis spectra, 3D scenes and mathematical graphs), `schermo()` (section, tab, controls with value and range, tables up to 4 × 40 rows, visible text), `immagine()` (visible canvases, `BSILab3D` scenes re-rendered, SVGs rasterised; JPEG at most 1400 px wide), `chiedi()` (image in the first turn only, conversation in memory; a box unchecked by the user, `ST.schermoNo`, is respected even when a named or catalogue section is opened: only the question); without a model it says so and shows what the app knows on its own; `bsi-ai-hub.js` — «🧠 Spiega» tab second after «💬 Chat», `window.bsiLLM`, per-provider image conversion; `bsi-spectra-figura.js` — the figure in the tab, «🧠 Spiega» button	`test_spectra_figura` — 14 checks per width (412 and 1280 px): second tab, «Lezione» absent; figure in the tab; with «Spettri Ref.» open the screen is read (≥ 1 graph, control values, image); five levels; the model receives screen text, question and image; exhaustive, non-simplified request with δ, multiplicity, J, integral and 3D scenes; mandatory structure; formatted answer and Spectra thinking; follow-up without re-sending the image; advanced PhD with frontier, open problems and primary literature; without a model it does not pretend; SVG spectra read as an image; 3D scene of «Meccanica orbitale 3D» (2 canvases); privacy: with «Includi lo schermo» unchecked, the named section opens but the model receives only the question, with no screen text or image. **Not tested**: the quality of a real model's explanation — the model's answers are simulated, the bench checks what is sent and how it is shown

3. Stability requirements

ID	Requirement	Implementation	Verification bench
STA-01	A long session must accumulate neither DOM nodes nor timers	section lifecycle management	`audit_stabilita` §1 — 194 sections × 5 rounds
STA-02	Exhausting `localStorage` must not make the app unusable	guarded writes; persistent warning in the File Manager	`audit_quota` (10), `test_filemanager` (15)
STA-03	No rejected promise may remain unhandled	systematic `.catch()`	`audit_promesse` — 22 checks, healthy network and dead network
STA-04	Corrupted saved data must not prevent startup	`loadJSON()` with fallback	`audit_stabilita` §4
STA-05	A slow network must not be treated as an absent one	`ATTESA_RETE_MS = 3500` in `sw.js`	`test_sw` §3 and §3-bis — measured 3,507 ms
STA-06	An update must not destroy work in progress	`BSI_SW_UPDATED` message, decision taken page-side	`test_sw` §4

ID	Requirement	Implementation	Verification bench
STA-07	Browsing molecules must not exhaust the WebGL contexts	reuse of the 3D viewer at all 5 creation points	`test_visore3d` — from 7 constructions down to 1
STA-08	An unavailable IndexedDB store must be declared, not fatal	`dbStore()` over a promise; persistent warning	`test_filemanager` §2

4. Interface and accessibility requirements

ID	Requirement	Implementation	Verification bench
UI-01	Every page must open without JavaScript errors	—	`audit_stabilita` §5 — 14 pages
UI-02	Charts must be sharp on high-density screens	`bsiNitido()` on the canvas contexts	`audit_grafici` — 40 canvases
UI-03	The app must work at 390 px width, and content that does not fit must stay **reachable**	`auto-fit` grids, no fixed column, and wide tables inside a container with `overflow-x:auto`	`audit_mobile` — **two distinct measurements** across all 194 sections at 390 px: the document's `scrollWidth` (the horizontal bar the user sees) **and** elements wider than their section **with no scrollable ancestor**, that is content clipped and unreachable — a written column that cannot be read and gives no sign of existing. **Both ways**: a wide table inside a scrolling container is **not** reported, because it can be read by scrolling. `audit_stabilita` for errors in a phone viewport
UI-04	The user must be able to know which version they are running and force the update	"Updates" entry in the ✨ panel	`test_aggiorna` — 9 checks
UI-05	Data deletion must be selective and reversible in its choices	`bsiCancellaDati()` by group	`browser_reset` — 24 checks
UI-08	Surfaces that appear only after an action must also meet WCAG AA contrast	the six panels of the Cheminformatics section	`test_cheminfo` — the analysis is run, the panels opened in turn, **460 text elements measured, 0 defects**, with the WCAG formulas rewritten inside the bench

ID	Requirement	Implementation	Verification bench
UI-09	Opening a section must not block the page	3Dmol's first `render()` moved out of the click; the 296 synthesis figures **and the 263 drug cards** drawn in 8 ms slices	`test_fluidita` — **194 sections traversed**, median switch 16 ms, worst 87 ms, **0 sections above 100 ms** (there were 3 with the new drugs, at 159 ms); plus the opposite proofs: the WebGL canvas appears anyway, all 296 figures exist shortly after, and printing does not come out mute. For the viewer the claim is **structural, not temporal**: immediately after the click handler returns, the WebGL context must not exist; shortly after it must — because a millisecond threshold was measuring SwiftShader rather than the code (14…192 ms on identical code)
UI-10	The interface must switch among many languages, declaring how much each is translated, and carrying the writing direction with it	`bsi-lingue.js` with **fourteen languages**, a searchable selector, `dir="rtl"` for Arabic	`test_lingue` — **170 skeleton elements out of 170 in each of the thirteen languages**, coverage computed by applying the dictionary and counting; writing direction tested in both directions; navigation categories and the search placeholder included
UI-11	The molecule must show how it holds together, and the viewer must choose the angle	`bsi-mol3d.js`: formation from dust with staggered arrival, and atom selection by click	`test_mol3d` — the canvas **pixels** at 130 ms and 2.3 s, the arrival order (skeleton before hydrogens) measured on each atom's progress, and the selection of 1, 2, 3 and 4 atoms
UI-12	Whoever has a file must be able to open it: a picker listing extensions does not narrow the choice on a phone, it **hides** it	`bsi-spettrolettore.js` — a single `<input type="file">` **with no `accept`**, a single button, and `apriQualunque()` routing by type, by the extension the format declares and **by content**, announcing the decision in `#bsiSP-deciso`	`test_spettrolettore` §7 — one visible button, an **empty** `accept` (a non-empty `accept` here *is* the defect), and routing verified on PDF, `.docx`, `.txt`, `.jdx`, two `.csv` columns and an image; **both ways**, a binary disguised as `.txt` is not poured out as text but declared binary
UI-13	The version shown in the header must be the one actually running	`index.html` — the badge is filled from `window.BSI_APP_VERSION`, no longer a hand-written string	`test_spettrolettore` §8 — the badge text is compared with `BSI_APP_VERSION` measured in the running page. It was needed: for about eighty versions the header read `v13l` when the application was already past version 190, and no check looked at it

5. Security and privacy requirements

ID	Requirement	Implementation	Verification bench
SEC-01	No API key may be present in the repository	keys only in `localStorage` or in the Worker's secrets	`tools/verifica-sicurezza.js` — 570 tracked files, 8 credential shapes
SEC-02	No personal data may leave the device without an explicit action	local-first architecture, telemetry disabled	`audit_rete` — **direct verification**: a canary value seeded into 71 stores of the user's data, the application used across 194 sections on 6 pages, and the URL, headers and body of every request inspected. Plus `verifica-sicurezza` on the two exit mechanisms
SEC-03	Passwords must not appear in clear text in the source	SHA-256 in `file_manager.html`	`tools/verifica-sicurezza.js`
SEC-07	No credential may EVER have entered the repository, not even in a commit later fixed	no key has ever been committed; keys live in `localStorage` or in the Worker's secrets	`audit_storia` — **1,551 distinct versions of text files across 390 commits**, 9 patterns tested in both directions; fails on a shallow clone, because it would measure less surface
SEC-04	No conflict marker may reach publication	—	`verifica_guida` §12 — 58 text files
SEC-08	The phrase with which the creator makes himself known to Spectra must not appear in the code nor be stored, and must not reach dialogue, memory, voice or model	`bsi-spectra-creatore.js` — SHA-256 fingerprint compared with the File Manager's; in proof mode the console field is a password field and sending skips `elabora()`; three wrong attempts, then a one-minute wait; in `localStorage` only `bsi-spectra-creatore-v1` with `ok`, `dal`, `ultima`, erasable through «Clear data → Persistent memory»	`test_spectra238` — «I am your creator» asks for proof and the field becomes a password field; a wrong phrase is rejected and the field goes back to normal; the typed phrase remains neither in the dialogue, nor in the page, nor in memory; after three attempts the wait; for the model the guest is not recognised, with the instruction never to ask for the phrase in chat; recognised: greeting by name, telemetry, context for the model; «forget that I am your creator» removes the recognition. `tools/verifica-sicurezza.js` — no plaintext password in the code

ID	Requirement	Implementation	Verification bench
SEC-09	The text Spectra speaks must leave the device only if the user has saved an OpenAI key, and the flow must be declared	`bsi-spectra-voce.js` — `parla()` calls `api.openai.com/v1/audio/speech` only if `chiaveOpenAI()` finds a key in `bsi_api_keys` and Web Audio is available; otherwise `speechSynthesis`. In Spectra Studio the 🔊/🔇 switch turns the voice off and the choice is kept in `bsi-spectra-studio-voce`: with the voice off no text goes to OpenAI. Declared in `03-Security-Privacy-Compliance.md` §1.2 and in `09-Release-Conformance-Statement.md`	`test_spectra_studio` — without a key `qualita()` is `'sistema'`, i.e. no request to OpenAI. **Not tested by the bench**: that with a key the text sent is only the text to be spoken (the bench does not reach OpenAI) and the studio's voice switch

> **SEC-03 — declared limitation.** GitHub Pages serves static files: any page-side access control can be bypassed by whoever reads the source. The File Manager's protection is a deterrent, **not** a security control, and the documentation says so. Furthermore the password remained in clear text in the git history until its removal, and taking a secret out of the files does not take it out of the history: `git log -p` hands it to anyone. The only effective remedy was to change it, and that **has been done** at version `bsi-v188`. The old one remains in the history and no longer opens anything.

> **SEC-08 — declared limitation.** The File Manager's limitation applies: the fingerprint is in the code and the recognition mark is in `localStorage`, where whoever holds the device can write it by hand. It is a deterrent, not an access control, and it protects no data: it tells Spectra whom it is talking to.

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6. Requirements not yet verified automatically

Declared for completeness. They are covered by documentary inspection or by manual testing, and their automation is planned.

ID	Requirement	Current coverage
UI-06	Full WCAG 2.1 AA conformance	the mechanical part is automated across **13 pages and 194 sections** (`tools/verifica-accessibilita.js`, 38,407 text elements: it was 19,751 before text over gradients entered the measurement, and 37,409 before the «Spectrum reader» section). **Of the 1,069 contrast defects that emerged, none remain: 0 measured** over those same 194 sections, and the value is recorded as a baseline the bench defends. Unlabelled fields are **zero**: see the box in `docs/09` §4. What remains outside is text inside SVGs and text over a real background image, counted on every run (D-09)
PERF-01	First-paint time on a low-end device	manual cross-device testing (`docs/02` §4)

ID	Requirement	Current coverage
SCI-11	Structures of 2 entries that are not single molecules (it was 6)	**not representable**: Ivermectin is a mixture of homologues, Coartem a combination of two active ingredients. The other four were closed by taking the structure from ChEMBL, see `docs/06` §2.4
PERF-02	Code coverage of the verification benches	**partially measured**: 49.89 % of statements over the widest path (`audit_copertura`). Whole-battery coverage and branch coverage remain unmeasured; see §8
UI-07	Behaviour on Firefox and WebKit	**not verified** — the battery runs on Chromium only; see §8
UI-21	Three apps in one: the ✨ menu, Spectra, Study and Utility must change with the subject, and the quick tools must work	**not automated** — `test_approfondimenti` checks only the bar's groups per subject (UI-18). The per-subject ✨ menu and the quick tools of `bsi-attrezzi.js` are tested by hand; their computations use `BSIMatCAS`, verified by `test_matematica` (SCI-41)
AI-13	Spectra Live: local commands without network, voice conversation, tools that read the tab and act on its controls	**not automated** — `bsi-spectra-live.js` is tested by hand. Speech recognition and synthesis are not available in the battery's headless Chromium

7. Overall coverage

Category	Requirements	Verified by a bench	Coverage
Scientific (SCI-01…59)	57	57	100 %
AI agent (AI-01…18)	17	17	100 %
Stability (STA-01…08)	8	8	100 %
Interface (UI-01…28)	25	25	100 %
Security (SEC-01…09)	7	7	100 %
Architecture (ING-01…02)	2	2	100 %
Automated total	**116**	**116**	**100 %**
Not automated (§6)	7	0	0 %
Declared total	**123**	**116**	**94 %**

Coverage is computed over the requirements **declared in this document** and must not be confused with code coverage: it measures how many requirements have a bench verifying them, not how many lines are executed during the tests.

> **Why two totals.** Reporting only the 100 % of automated requirements would be true and misleading at once: it is 100 % of what was *chosen* to be automated. The number that matters to an assessor is the second one — **116 requirements verified out of 123 declared** — and the seven that are missing are listed by name in §6, not summarised into a percentage.
>
> This table is checked by `tools/verifica-documenti.js`, which counts the identifiers actually present in the document and fails if the totals written here do not match. It is a check born of a real error: the totals said 36 over 9 scientific categories while the identifiers defined were 10, and `SCI-10` appeared twice — once as a verified requirement and once as an unverified one.

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8. Declared coverage gaps

These are not defects: they are things that have **not been measured**, and that therefore cannot be asserted.

Gap	What it entails	Why it is so
Partial code coverage	Measured: **49.89 %** of statements over the widest path a bench walks. It is not the coverage of the whole battery, and it is statements, not branches	Chromium's profiler collects it inside the engine, with no build and without rewriting the source: the obstacle belonged to the tool (c8, istanbul), not to the problem. Value recorded; `audit_copertura` fails if it falls
Chromium only	Behaviour on Firefox and WebKit is verified by hand, not by a bench	The battery uses `playwright-core`, which downloads a single engine. In the verification environment the CDN for the other engines answers **403** to the network policy: they cannot be installed there
No visual regression	An unintended graphical change would not be caught	Spectra are deterministic and comparable byte for byte (SCI-05): the comparison exists on the traces, not on the whole page
2 entries without a structure	Two entries out of 178 have no structure to show: Ivermectin is a mixture of homologues, Coartem a combination of two active ingredients. It was six	See `06-Scientific-Accuracy-Data-Provenance.md` §2.4
Performance on slow devices	First-paint time is not measured on low-end hardware	It requires physical devices; the test is manual

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Document updated to version `bsi-v241`.